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# Efficient Catalysis for Zinc–Air Batteries by Multiwalled Carbon Nanotubes-Crosslinked Carbon Dodecahedra Embedded with Co–Fe Nanoparticles

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**The design and fabrication of nanocatalysts with high accessibility and sintering resistance remain significant challenges in heterogeneous electrocatalysis. Herein, a novel catalyst is introduced that combines electronic pumping with alloy crystal facet engineering. At the nanoscale, the electronic pump leverages the chemical potential difference to drive electron migration from one region to another, separating and transferring electron-hole pairs. This mechanism accelerates the reaction kinetics and improves the reaction rate. The interface electronic structure optimization enables the CoFe/carbon nanotube (CNT) catalyst to exhibit outstanding oxygen reduction reaction (ORR) and oxygen evolution reaction (OER) performance. Specifically, this catalyst achieves an ORR half-wave potential ($E_1/_2$) of 0.895 V, outperforming standard Pt/C and $RuO_2$ electrocatalysts in terms of both specific activity and stability. It also demonstrates excellent electrochemical performance for OER, with an overpotential of only 287 mV at a current density of 10 mA $cm^{-2}$. Theoretical calculations reveal that the carefully designed crystal facets reduce the energy barrier of the rate-determining steps for both ORR and OER, optimizing $O_2$ adsorption and promoting the oxygen capture process. This study highlights the potential of developing cost-effective bifunctional ORR–OER electrocatalysts, offering a promising strategy for advancing Zn–air battery technology.**

## 1. Introduction

In recent years, zinc–air batteries (ZABs) have garnered considerable attention as potentially inexhaustible and sustainable green energy conversion systems, owing to the abundant availability and eco-friendliness of zinc.[1–3] Over the past decade, ZABs have been extensively studied and rapidly developed. Compared with solar and wind energy, which are highly dependent on climatic conditions, electrochemical technology has emerged as a key solution to energy challenges.[4–7] ZABs are promising energy storage solutions because of their outstanding rechargeability and high energy density. Various electrocatalysts have been developed to address the slow kinetics and high potentials related to the oxygen reduction reaction (ORR) and oxygen evolution reaction (OER) during the charge–discharge cycles in ZABs.[8–10] To date, commercial precious metals such as Pt/C have shown high activity in the ORR, whereas $IrO_2/RuO_2$

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is optimal for the OER.[11–13] Nevertheless, the high cost and limited storage capacity of precious metal materials restrict their extensive commercial use. Therefore, it is crucial to develop cost-effective and efficient reversible oxygen catalysts as substitutes for precious metal catalysts.[14–16]

In recent years, metal-nitrogen-carbon (M-N-C) materials featuring biomimetic porphyrin structures have demonstrated Pt-like activity in catalyzing the ORR, showing significant promise to researchers. Structures such as Fe−N−C, Co−N−C, Cu−N−C, and Mn−N−C have been advantageous for ORR.[17] However, for ORR catalysts to be effective, they must comply with Sabatier's principle to attain a suitable M−O binding strength. Considering the performance differences between various materials and Pt-based catalysts in ORR,[18] further refinement of these potential structures is necessary to adjust their electronic properties, thereby promoting an efficient ORR.[19–22] Heteroatom doping is an effective strategy; for example, introducing heteroatoms with varying electronegativities compared with those of N or C can modify the electronic properties of adjacent atoms. For instance, He et al. demonstrated that doping oxygen atoms into $FeN_4$-NCR enhances the ORR activity by lowering the d-band center of Fe. However, most developed catalysts primarily enhance the ORR performance. Addressing the challenge of improving OER activity to construct rechargeable ZABs remains difficult.[23–26] Additionally, the low solubility of $O_2$ in water severely affects the ORR owing to the mass-transfer limitations. Thus, the morphology of the catalyst plays a crucial role in the ORR. Therefore, increasing the specific surface area of catalyst materials is essential for achieving higher unit activity.[27,28] Furthermore, Liu et al. embedded Fe single atoms/clusters onto nitrogen-doped carbon, achieving half-wave potentials of 0.71 V in acidic environments and 0.90 V in alkaline environments. Given that previous material designs have not achieved bifunctional activity for both the ORR and OER, it is crucial to redesign material synthesis and functionality to endow catalysts with bifunctional activity.[29–31]

Based on these findings, we developed a novel CoFe/CNT nanocomposite catalyst. This innovative catalyst was synthesized by the in situ growth of multiwalled CNTs (MWCNTs) with a core–shell structure on CoFe bimetallic nanoparticles anchored on ZIF-8 dodecahedra. The catalyst was synthesized from a CoFe/ZIF-8 precursor via a one-step high-temperature carbonization process, resulting in carbon-coated CoFe alloy nanoparticles. During synthesis, Fe, Zn, and Co ions were simultaneously introduced into the 2-methylimidazole solution. Because of the molecular polarity effects, the lone-pair electrons on the outermost N atoms of 2-methylimidazole interacted with the 3d vacant orbitals of Zn atoms, leading to chemical precipitation and the formation of ZIF-8-Co/Fe nanododecahedra. These particles were then mechanically ground into a fine powder and carbonized to prepare the precursor, ultimately yielding a CoFe/CNT nanocomposite, as shown in **Figure 1a**. This study introduces a mechanism that combines electronic pumping with alloy crystal facet engineering. At the nanoscale, electronic pumping utilizes the chemical potential difference to drive electrons from one region to another, separating and transferring electron–hole pairs, which accelerates reaction kinetics and improves reaction rates. When applied to alloy materials, particularly those with varying exposed crystal facets, carefully engineered facets can optimize the electron transport pathway, thus enhancing the catalytic performance. In the ORR, the catalyst achieves an $E_{1/2}$ of 0.895 V, whereas, in the OER, it delivers a low overpotential of 287 mV at a current density of 10 mA $cm^{-2}$. This strategy of optimizing heterostructures to promote charge transfer provides a guiding pathway for the development of low-cost, efficient bifunctional ORR–OER electrocatalysts and holds promise for advancing ZAB technology.

## 2. Results and Discussion

### 2.1. Synthesis and Characterization of Materials

To develop bifunctional catalysts conducive to enhancing both ORR and OER, strategies such as heteroatom doping and morphology control have been utilized. The introduction of Fe metal to form CoFe alloy particles significantly improved their stability. Additionally, the in-situ-grown MWCNT structure led to a significant increase in the specific bifunctional performance. Initially, Fe, Zn, and Co ions were introduced simultaneously into a 2-methylimidazole solution. Owing to the polar effects of the molecules, the lone pair of electrons on the outermost N atoms entered the 3d empty orbitals of the Zn atoms, forming ZIF-8-Co/Fe nanododecahedra through chemical precipitation. The structure was mechanically milled into a fine powder and carbonized to prepare the precursor, yielding CoFe/CNTs, as illustrated in Figure 1a. During pyrolysis, the Fe–Co alloy was embedded within the carbon matrix derived from 2-methylimidazole, leading to the spontaneous formation of CNT structures. The resultant catalyst demonstrates excellent bifunctional activities for both ORR and OER, with an ORR half-wave potential of 0.895 V and an overpotential of 287 mV at a current density of 10 mA $cm^{-2}$. This approach to optimizing heterostructures for enhanced charge transfer offers a promising strategy for preparing low-cost and straightforward bifunctional ORR + OER electrocatalysts.

The surface morphologies and crystalline structures of the prepared electrocatalysts were characterized using scanning electron microscopy (SEM), transmission electron microscopy (TEM), and TEM–energy dispersive X-ray spectroscopy (TEM–EDS). The SEM image of CoFe/CNT in Figure 1b shows that after the high-temperature treatment, the surface of the electrocatalyst was rough, with an average nanoparticle diameter of ≈200 nm. The TEM image in Figure 1c shows a distinct hollow box-like structure with a thin shell layer. Such a hollow box structure may facilitate an increased contact area between the electrolyte and the catalyst, enhancing the number of active sites and promoting the transfer of reaction intermediates. As shown in Figure S1 (Supporting Information), numerous CNTs internally encapsulating metal particles were grown in situ on the surface of the nanododecahedron. The image shows the core–crown-structured MWCNTs encircling the CoFe bimetallic nanoparticles anchored on the ZIF-8 nanododecahedron. The formation of nanotubes is not only uniform but also closely integrated with the CoFe alloy nanoparticles, substantiating the validity of our design concept and synthesis methodology. This growth may be induced by adjacent metal particles that catalyze the growth of the CNTs. These nanostructures can accelerate charge transfer and possess a large surface area to enhance catalyst utilization efficiency, thus achieving the performance required for efficient

 

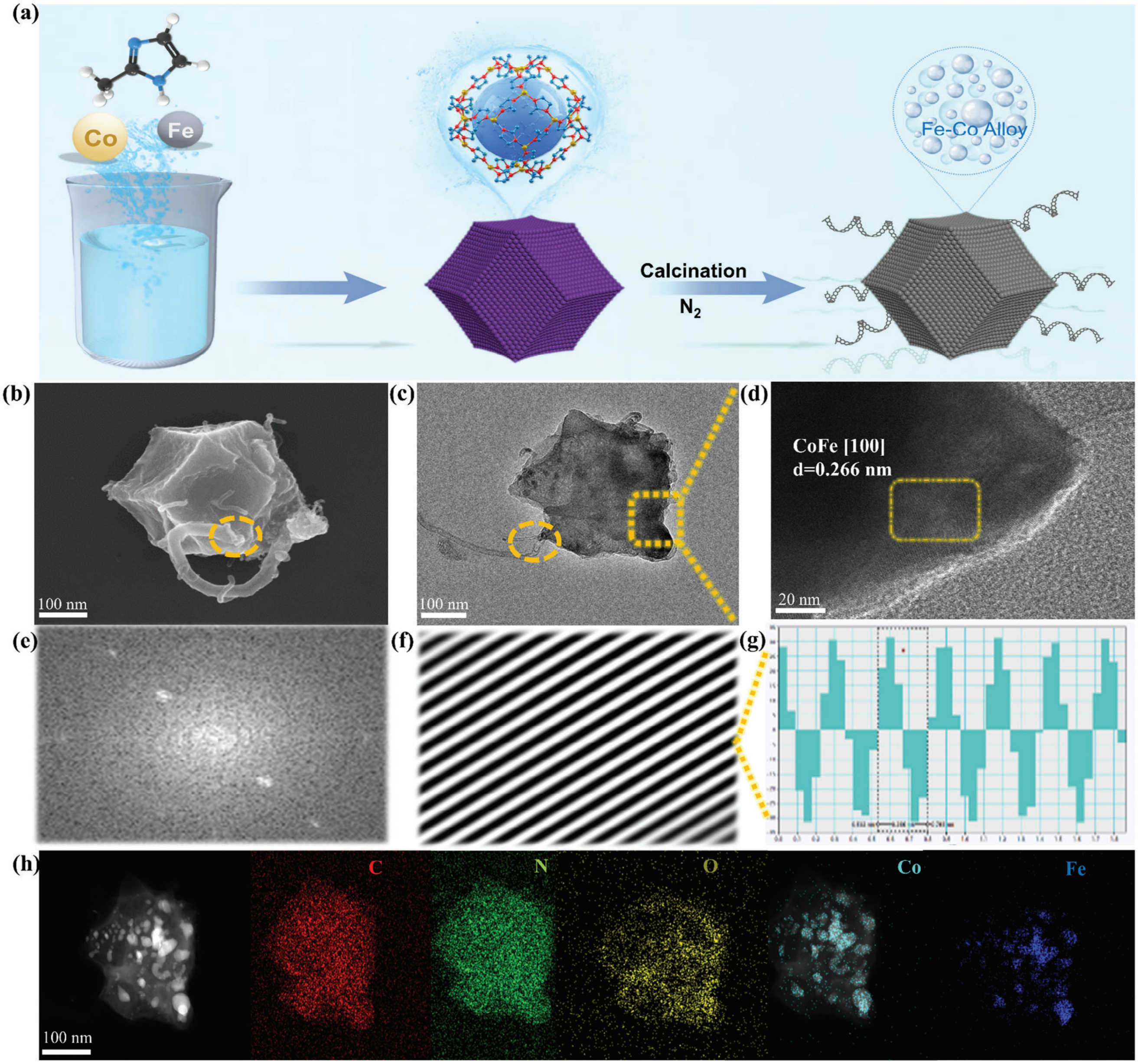


Figure 1. CoFe/CNT nanocomposite: a) Schematic illustration of the synthesis process; b) a SEM image; c) TEM image; d–g) high-resolution TEM images (HRTEM); h) energy dispersive X-ray spectroscopy (EDS) elemental distribution maps.

ZABs. Furthermore, after the ORR stability reaction, the morphology of CoFe/CNT remained largely unchanged, as shown in Figure S2a,b (Supporting Information), with the form and presence of the CoFe alloy species being virtually unaltered. This stability may explain why CoFe/CNT maintained its ORR performance. In addition, NC, CoNC, and FeNC exhibited dodecahedral ZIF-8 structures, as shown in Figure S3a–c (Supporting Information). However, only CoFe/CNT demonstrated the in situ growth of MWCNTs on CoFe/ZIF-8 composite materials, forming a distinctive core–crown structure, as shown in Figure S3d (Supporting Information). An interlayer spacing of ≈0.266 nm visible in Figure 1d, corresponds to the (100) plane of graphitic carbon, further confirming the formation of high-temperature carbonized graphite. Furthermore, high-resolution lattice fringes obtained through reverse Fourier transformation are shown in Figure 1e–g, indicating successful embedding of the Fe–Co alloy into the carbon framework. Additionally, the highly graphitized carbon framework not only improves the conductivity of the electrocatalyst, ensuring efficient and stable charge transfer but also protects and maintains the nanoparticles within the carbon framework in an appropriate state, thereby significantly enhancing the electrocatalytic performance for oxygen. Optical photographs of CoFe/CNT and its precursor, shown in Figure S4a,b (Supporting Information), reveal that the precursor is a pinkish powder with a metallic Co color that turns into a black carbon-based solid after

 

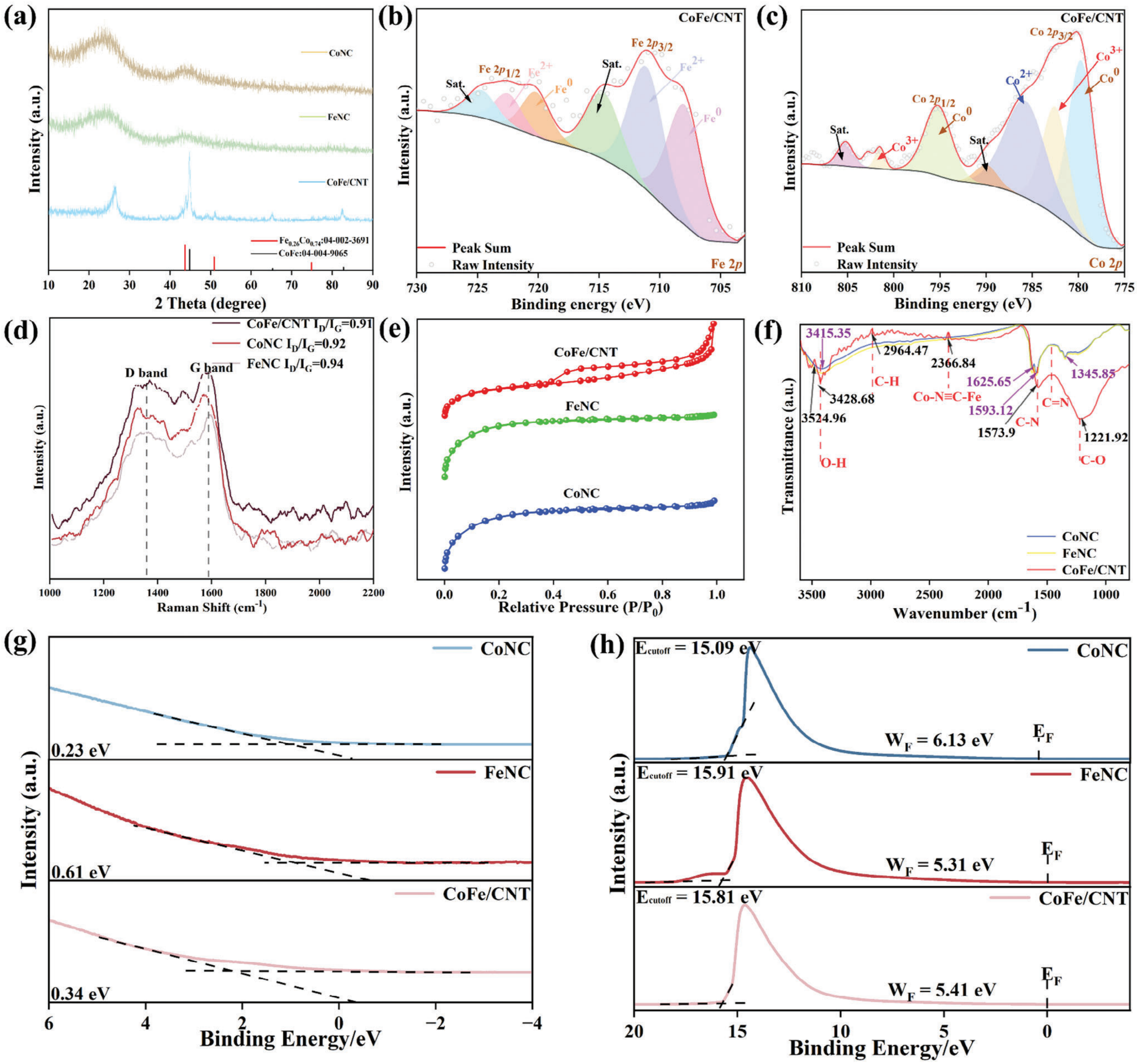


**Figure 2.** Characterization and performance analysis of CoFe/CNT: a) X-ray diffraction (XRD) patterns; b) Co 2p X-ray photoelectron spectroscopy (XPS) spectrum of CoFe/CNT; c) Fe 2p XPS spectrum of CoFe/CNT; d) comparison of Raman spectra of different materials; e) nitrogen desorption/adsorption isotherms of CoFe/CNT, CoNC, and FeNC; f) Fourier transform infrared (FTIR) spectra of CoFe/CNT, CoNC, and FeNC powders; g) valence band maximum values; h) working functions of the as-prepared catalysts.

high-temperature calcination. The TEM–EDS elemental mapping images revealed strong signals for C, N, O, Fe, and Co, as illustrated in Figure 1h. At the scale of 100 nm in the aberration corrected scanning transmission electron microscopy (AC-STEM) image, numerous bright spots in the atomic size distribution were observed. Because AC-STEM imaging utilizes the high-angle annular dark-field (HAADF) mode, bright spots at the atomic level correspond to metal elements with high atomic numbers. In CoFe/CNT, given the presence of only two metal elements, the bright white spots in the AC-STEM image are likely alloy particles of CoFe, further suggesting that both the Fe-Co alloy and CNT structures were uniformly distributed throughout the carbon framework.

The XRD results shown in **Figure 2a** indicate that the introduction of CoFe nanoparticles as active catalysts resulted in additional diffraction peaks in the XRD spectra, confirming the successful synthesis of the CoFe alloy phase on a carbon substrate. These new peaks, located at ≈2$\theta$ values of 44.7 and 65.2°, were attributed to the (100) planes of the body-centered cubic structure of the CoFe alloy, suggesting that introducing the catalyst affects the microstructure of the electrode materials and possibly enhances the electrochemical performance by providing more active sites. The CoFe/CNT material, comprising two alloy phases of FeCo, further confirmed that the final product was a nanododecahedron embedded in a CoFe alloy. The chemical composition of the CoFe/CNTs in the nanocomposite material was characterized using XPS. Figure 2b,c shows the Co 2p and Fe 2p core-level spectra, respectively, obtained by curve fitting. The Co 2p peaks at binding energies of 779.1 and 793.3 eV correspond to $Co^0$ in CoFe/CNT, whereas the similar results for Fe 2p peaks at 706.7 and 720.1 eV correspond to $Fe^0$. Additionally, the presence of $Fe^{2+}$ and $Co^{2+}$ indicates surface oxidation of the alloy particles. The coexistence of $Fe^{2+}$ and $Fe^{3+}$ suggests complex surface oxidation reactions, likely directly related to the catalytic activity of the electrode, further confirming the presence of the CoFe alloy within the compound system. By contrast, the spectra for Co 2p in Figure 2c,d reveals the presence of cobalt in the forms of $Co^{2+}$ and $Co^{3+}$. This multivalent state of Co enhances the electrochemical activity of the material and potentially improves the electron transfer efficiency during the electrode processes.[32–34] Moreover, the XPS spectra of the CoFe/CNT samples before and after use demonstrated that the phase structure and chemical composition of the catalyst demonstrated considerable consistency before and after the electrocatalytic reaction, suggesting that the modified catalyst exhibited significant chemical stability, as shown in Figures S5a–f (Supporting Information). Raman spectroscopy, Fourier transform infrared (FTIR), and electrochemical testing were employed to comprehensively evaluate the surface characteristics and electrochemical behavior of the ZAB electrode materials.[35,36] In the spectral range of 1000 to 2000 $cm^{-1}$, the Raman spectrum of the sample exhibits two prominent peaks at ≈1360 and 1590 $cm^{-1}$,[37] corresponding to the D and G bands, respectively. The intensity ratio of the D band to the G band ($I_D/I_G$) is a key indicator of the graphitization level of a material,[38] where a lower $I_D/I_G$ ratio suggests a higher degree of graphitization. The calculated $I_D/I_G$ ratio of the CoFe/CNT sample was 0.91, indicating a high level of graphitization, which enhanced electrical conductivity and facilitated electron transfer during the ORR and OER. This high degree of graphitization contributed to the superior electrocatalytic performance of the CoFe/CNT sample compared with that of the other materials analyzed, as shown in Figure 2d, and facilitated faster electron transfer during the ORR and OER, thereby enhancing the electrocatalytic performance of the material.[39–41] These shifts suggest that the hydrophilicity and electronic structure of the electrode materials have been optimized, enhancing the electrolyte contact and reaction kinetics.

As shown in Figure 2e, the specific surface areas and pore structures of the CoNC, FeNC, and CoFe/CNT were evaluated using the nitrogen adsorption/desorption method. The noticeable hysteresis loop in CoFe/CNT can be explained by the Kelvin equation, which suggests that, in porous adsorbents, a concave meniscus formed during the initial stages of adsorption results in vapor pressure over the meniscus that is always less than the saturated vapor pressure over a flat liquid surface.[42] Consequently, vapor condensation occurs at pressures below the saturation vapor pressure, typically moving from smaller to larger pores as the gas pressure increases. During desorption, the curvature radius of the liquid surface after capillary condensation is always smaller than that before condensation, resulting in a desorption pressure that is lower than the adsorption pressure for the same amount of adsorption. Although the specific surface area of CoFe/CNT is 537.4 $m^2$ $g^{-1}$, which is less than that of the comparative materials CoNC and FeNC, the introduction of Fe metal to form CoFe alloy particles significantly enhanced the stability and performance of the material. These alloy particles might cause some agglomeration, reducing the specific surface area; however, their alloy structure aids in enhancing the catalytic activity. Additionally, the formation of CNTs may have caused the collapse or blockage of the pore structure, reducing the specific surface area.[11,43,44] However, their excellent conductivity and unique structural characteristics significantly enhance their bifunctional performance. The conditions used during mechanical milling and carbonization, while adversely affecting the pore structure, also optimize the electrochemical performance of the material. Furthermore, although the heteroatom doping strategy altered the pore structure and surface properties, it further enhanced the catalytic activity and stability of the material. Overall, these factors collectively led to a significant improvement in performance despite a reduction in the specific surface area of the final sample. FTIR analysis, as shown in Figure 2f, further revealed changes in the chemical bond structures of the electrode materials after the reaction, particularly in the absorption peak intensities of the C—O and C—N bonds, indicating the processes of surface oxidation and nitrogen doping. The absorption peak of the C—O bond shifted from 1040 to 1075 $cm^{-1}$, and the C—N bond intensified from 1380 to 1415 $cm^{-1}$.[45]

The maximum valence band value represents the highest energy level that electrons can occupy in the valence band of the material. As shown in Figure 2g, the maximum valence band values of the different materials vary, reflecting the differences in their electron mobilities. For instance, CoFe/CNT has a relatively low valence band maximum (0.34 eV), indicating that electrons can be more easily excited to the conduction band. This lower valence band maximum contributes to enhanced electron transfer efficiency, thereby improving the catalytic performance of the material. Specifically, this facilitates the transfer of electrons from the CoFe nanoalloy to the reaction surface, which is critical for both ORR and OER. By contrast, FeNC and CoNC exhibit maximum valence band values of 0.61 and 0.23 eV, respectively. Although CoNC has a lower maximum valence band value, its structure does not expose sufficient active sites or promote electron transfer as efficiently as CoFe/CNT, resulting in a comparatively lower ORR and OER performance. From the perspective of electron transfer, the integration of CoFe alloy nanoparticles with CNT reduces the barriers to electron transfer, allowing electrons to flow more easily from active sites to reactants, thereby accelerating the catalytic reactions. This improvement in electron transfer capability is further reflected in the high reaction efficiency of CoFe/CNTs, even at lower overpotentials. The Fermi level ($W_F$) is the highest energy level occupied by electrons in a material under equilibrium. Figure 2h shows that the Fermi level of CoFe/CNT is 5.41 eV. Although FeNC has a lower Fermi level, it exposes significantly fewer active sites during the reaction process compared with CoFe/CNT. The charge transfer model of interaction is shown in Figure S6 (Supporting Information). The nanoalloy structure formed between cobalt and iron facilitates electron redistribution, resulting in the creation of an internal electric field. This electric field drives the electrons from Fe (which has a lower work function) to Co (which has a higher

 

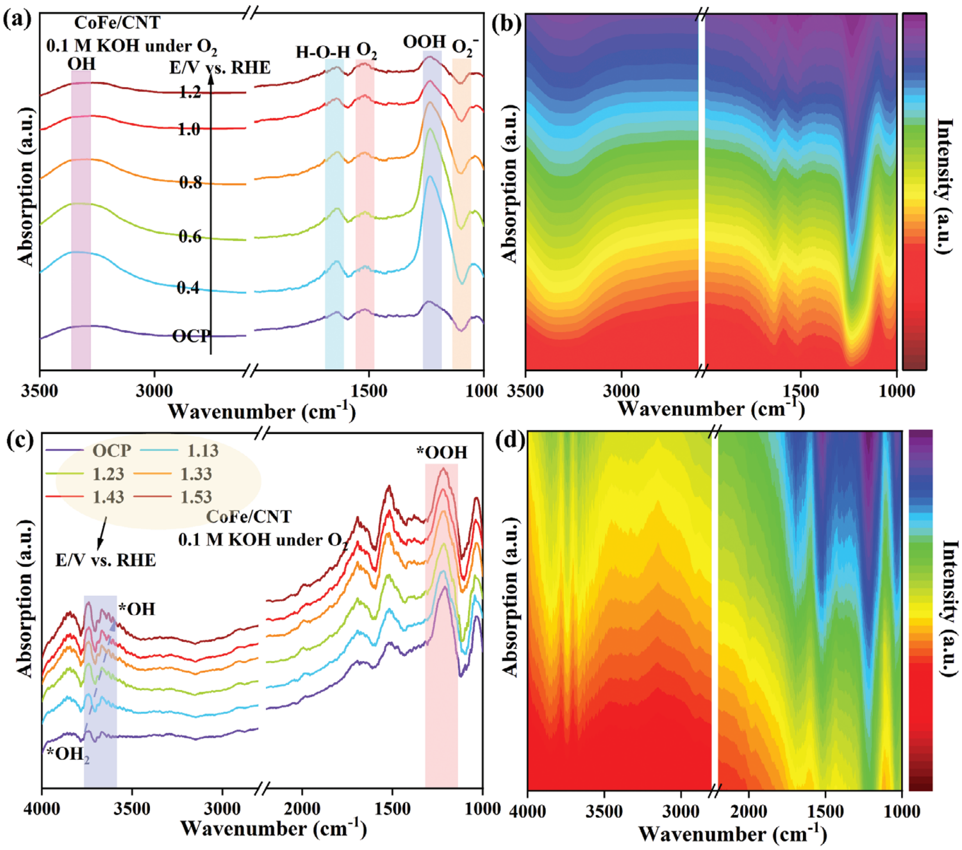


**Figure 3.** In situ attenuated total reflection surface-enhanced infrared absorption spectroscopy (ATR-SEIRAS) data of the CoFe/CNT catalyst during the ORR and OER in 0.1 м KOH solution. For ORR: a) ATR-SEIRAS spectra, and b) 2D heat map; for OER: c) ATR-SEIRAS spectra, and d) a 2D heat map.

work function), corresponding to the XPS results (Table S1, Supporting Information). During the formation of the FeCo alloy, an internal electric field is generated, resulting in the redistribution of charges within the material, subsequently driving the electron flow from Fe to Co. This redistribution optimizes electron transfer pathways, facilitating electron transmission through the active sites during catalytic reactions. The enhanced electron transfer promotes the adsorption and activation of reaction intermediates, particularly in the OER, where key intermediates such as $O_2^-$ and OOH are more readily activated. The internal electric field not only improves the electron transfer capabilities of the material in the OER but also lowers the reaction energy barrier, thereby accelerating the adsorption and conversion of oxygen intermediates. Notably, in the CoFe/CNT system, the electron flow from Fe to Co increases the charge density at the reaction sites, significantly optimizing the electron transport efficiency during catalysis and accelerating the overall OER.

**Figure 3** illustrates the in situ attenuated total reflection surface-enhanced infrared absorption spectroscopy (ATR-SEIRAS) data for the CoFe/CNT catalysts in 0.1 м KOH solution during the ORR and OER. This analysis aimed to elucidate the formation and evolution of reaction intermediates on the catalyst surface, thereby demonstrating the superior performance of the material. Figure 3a,b presents the ATR-SEIRAS spectra and the corresponding 2D heat maps for the ORR process. As the potential increased, the absorption peaks at 3500 and 1650 $cm^{-1}$ intensified significantly, indicating the active vibrations of the O—H and H—O—H bonds in the water molecules on the catalyst surface.[46,47] Concurrently, the absorption peaks at 1235 and 1120 $cm^{-1}$ gradually appeared and intensified, reflecting the formation and accumulation of *OOH and $*O_2^-$ intermediates, which highlights the high efficiency of the catalyst in promoting the ORR.[48–51] Figure 3c,d shows the ATR-SEIRAS spectra and 2D heat maps of the OER process. Similarly, as the potential increased, the absorption peaks at 3500 and 1650 $cm^{-1}$ notably intensified, signifying the critical role of water molecule adsorption and activation during OER.[52,53] Moreover, the emergence and enhancement of the absorption peak at 1235 $cm^{-1}$ further confirmed the pivotal role of *OOH intermediates in the OER process.[54–56] The in situ infrared image of CoNC is shown in Figure S7a,b (Supporting Information), and no significant change can be observed from the images. The ATR-SEIRAS data indicate that the CoFe/CNT catalyst demonstrates significant intermediate formation and adsorption activity in the ORR, as well as outstanding catalytic performance in the OER. These findings validate the efficiency and stability of the material in electrochemical reactions, highlighting its substantial potential for practical applications. This in situ infrared analysis further substantiates the innovative structural design of the CoFe/CNT catalyst and its superior performance in ZAB technology.

 

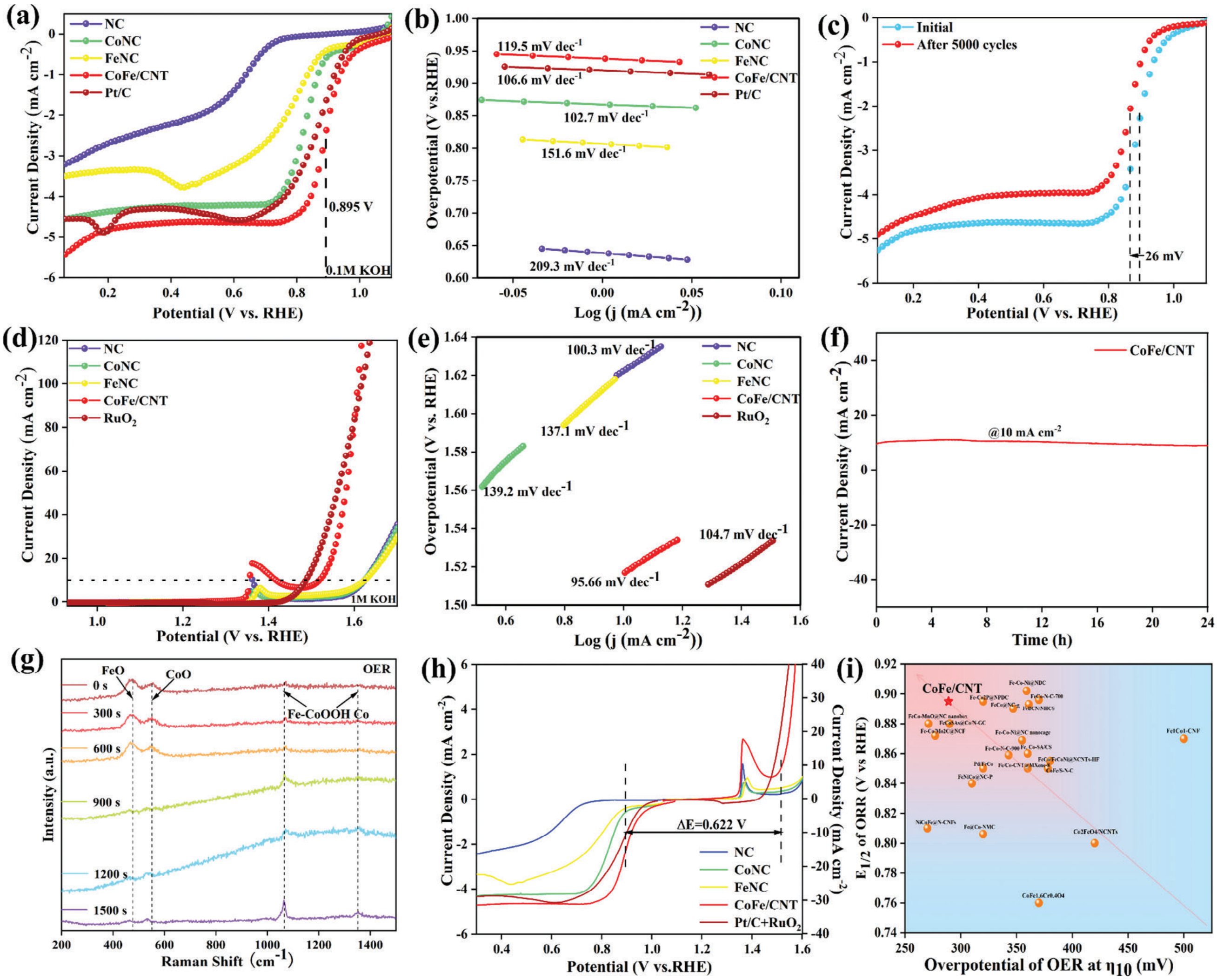

**Figure 4.** ORR activity of the studied catalyst: a) linear sweep voltammetry (LSV) curves obtained in a saturated 0.1 M KOH solution using different catalysts as working electrodes; b) Tafel plot; c) cycle stability of CoFe/CNT catalyzing ORR. OER activity of the studied catalyst; d) LSV curve recorded in 1 M KOH; e) Tafel plot; f) chronoamperometric *i*–*t* curve for OER. g) Raman spectra of CoFe/CNT at different reaction times during the OER; h) voltage gaps between the ORR and OER on different materials; i) comparison of OER performance of CoFe/CNT with that of recently reported non-precious metal catalysts.

### 2.2. ORR and OER Performance

Additionally, the aforementioned samples underwent linear sweep voltammetry (LSV) testing at a rotation speed of 1600 rpm in a saturated 0.1 M KOH solution, revealing that CoFe/CNT achieves a higher half-wave potential and limiting current. In **Figure 4a**, CoFe/CNT shows the optimal half-wave potential ($E_{1/2}$ = 0.895 V) and the highest limiting diffusion current density ($J_L$ = 5.45 mA cm$^{-2}$), surpassing the performance of the commercial Pt/C (20 wt.%) catalyst. Conversely, the ORR driven by CoNC ($E_{1/2}$ = 0.832 V) only reaches the performance level of the Pt/C catalyst, compared with RHE at 0.846 V, while the ORR $E_{1/2}$ with FeNC as the catalyst compared with RHE is only 0.795 V. Regarding limit current density ($J_L$), the ORR driven by CoFe/CNT outperforms the Pt/C catalyst ($J_L$ = 5 mA cm$^{-2}$). Conversely, CoNC ($J_L$ = 4.58 mA cm$^{-2}$) and FeNC ($J_L$ = 3.77 mA cm$^{-2}$) perform significantly worse compared with the Pt/C catalyst, likely owing to morphological factors. The CNT structure generated by the in situ growth in CoFe/CNT led to changes in the catalyst structure, exposing more active sites and increasing the reaction frequency, resulting in an increase in $J_L$. As shown in Figure S8a–e (Supporting Information), during the ORR process, the Cdl of CoFe/CNT is 7.74 mF cm$^{-2}$, markedly superior to that of other materials, and achieving the highest electron transfer rate. This confirms that the synergistic interaction between the Fe and Co alloy components and the CNT structure enhances the activity of the material. Additionally, LSV curves were collected at various rotation speeds, as shown in Figure S9a (Supporting Information), to investigate the ORR mechanism of the CoFe/CNT catalyst. As the rotation speed increased, the limiting current density ($J_L$) consistently increased, indicating that the occurrence of the ORR on CoFe/CNT was influenced by mass transfer processes.[57]

 

Notably, in the non-faradaic region, the initial current density for the ORR was slightly below zero, which was consistent with other reference results. This was attributed to the porous carbon matrix of the catalyst material, which exhibited capacitive effects. According to the Koutecky–Levich equation, the calculated number of electron transfers is ≈$n \approx 3.9$.[58] This suggests that $O_2$ undergoes a four-electron transfer electrochemical conversion to $H_2O$ on CoFe/CNT, as shown in Figure S9b (Supporting Information). The ORR mechanism was further investigated from the Tafel slope(Figure 4b), where the Tafel slope of CoFe/CNT is 119.5 mV $dec^{-1}$, slightly inferior to that of Pt/C. This may be because of the spontaneously generated CNT structure, which causes a higher surface roughness, and the alloy particles formed by the CoFe alloy, which result in higher material crystallinity, thereby increasing the limiting current and decreasing the electron transfer rate. Moreover, even after 5000 cycles, the $E_{1/2}$ for ORR catalyzed by CoFe/CNT decreased by only 26 mV (cathodic potential), as shown in Figure 4c. To further examine the catalytic performance of CoFe/CNT for the ORR, methanol tolerance and long-term durability are crucial parameters for evaluating both the catalyst performance and practical applications. As shown in Figure S10 (Supporting Information), after the introduction of 1 mL of methanol into the electrolyte cell, the prepared CoFe/CNT exhibited commendable methanol resistance. By contrast, the current density of Pt/C decreased significantly after the addition of methanol. These results demonstrate that CoFe/CNT possesses a robust methanol poisoning resistance, which makes it advantageous as a cathode catalyst for fuel cells.[59,60]

The CoFe/CNT sample also exhibited efficient electrocatalytic OER performance, which is crucial for the construction of high-performance rechargeable ZABs. As illustrated in Figure 4d, the derived CoFe/CNT in a 1 м KOH solution demonstrated excellent OER activity, with an overpotential of 287 mV at a current density of 10 mA $cm^{-2}$. By contrast, the OER performance of the FeNC ($\eta_{1n}$ = 392 mV) and CoNC ($\eta_{1n}$ = 394 mV) catalysts was poorer, whereas the OER performance of NC was negligible.[61–63] Compared with the commercial precious metal $RuO_2$ catalyst ($\eta_{1n}$ = 258 mV), CoFe/CNT exhibited more competitive performance, particularly at high current densities. From Figure S11a–e (Supporting Information), during the OER process, the Cdl of CoFe/CNT was 6.5 mF $cm^{-2}$, markedly superior to that of other materials, achieving the highest electron transfer rate. This further proves that the synergistic interaction between the Fe and Co alloy components and the CNT structure enhances the activity of the material.[64] The OER catalytic kinetics of the electrocatalyst were further investigated using Tafel slopes and electrochemical impedance spectroscopy (EIS). The OER mechanism was further investigated from the Tafel slope, as shown in Figure 4e, where the Tafel slope of CoFe/CNT was 95.66 mV $dec^{-1}$, better than that of $RuO_2$, 104.7 mV $dec^{-1}$, and other comparative materials. The corresponding EIS results are shown in Figure S12b (Supporting Information), which further demonstrate that the CoFe/CNT sample has the lowest charge-transfer resistance compared with the other electrocatalysts. Chronoamperometric measurements indicated that CoFe/CNT exhibited high OER stability with essentially no change in current after continuous operation for 24 h, as shown in Figure 4f. Moreover, even after 10 000 cycles, the OER activity of the CoFe/CNT catalyst showed a significant improvement, as shown in Figure S13 (Supporting Information). The ORR and OER kinetics of the electrocatalyst were further investigated using EIS.[65] In the high-frequency region, shown in Figure S12a (Supporting Information), CoFe/CNT exhibited the smallest semicircle diameter during the ORR, indicating the lowest charge-transfer resistance and highest efficiency. In the low-frequency region, shown in Figure S11b (Supporting Information), CoFe/CNT showed a more gradual increase in impedance during the OER, demonstrating excellent electrochemical performance and low charge transfer impedance. This confirms the superior performance of CoFe/CNT in the redox reactions.

Initially, we used Raman spectroscopy to explore the surface chemistry and structural dynamics of the FeCo/CNT materials during the OER, as shown in Figure 4g. By collecting data at different time points (0–1500 s) during the OER, we detailed the evolution of the Raman peaks of iron and cobalt oxides (FeO and CoO) and their mixed hydroxides (Fe-CoOOH). At the initial moment (0 s), we observed characteristic peaks of FeO and CoO at ≈600 and 950 $cm^{-1}$, corresponding to those reported in the literature for iron and cobalt oxides. Over time, the intensity changes and shifts of these peaks reflect the transformation of Fe and Co oxides in the electrochemical environment. By 1500 s, we particularly noted a significant enhancement of the Fe-CoOOH peaks at ≈300 and 600 $cm^{-1}$, indicating the formation of active Fe—Co mixed hydroxide phases during the catalytic process. Moreover, the formation of Fe-CoOOH was directly linked to the adsorption of intermediates on the catalyst surface and alterations in the electronic structure.[66] Time-dependent analysis of the Raman spectra shows that the enhancement of Fe-CoOOH is closely related to increased OER activity, supporting the view that FeCo/CNT materials enhance the catalytic efficiency by forming optimized active phases on their surfaces.[67] Thus, during the OER process, as the reaction progressed, the formation of active Fe—Co mixed hydroxide phases increased, thereby continuously improving the OER performance. Moreover, the important oxygen electrode activity parameter $\Delta E$ ($E_{j=10} \approx E_{1/2}$) for the CoFe/CNT catalyst was calculated, showing a minimum $\Delta E$ of 0.622 V as shown in Figure 4h, which is far superior to that of the other four comparative samples and other reported electrocatalysts. This demonstrates its exceptional bifunctional electrocatalytic performance. These properties make CoFe/CNT a beneficial catalyst for ZABs, exhibiting outstanding performance among similar catalysts. Although the OER performance of CoFe/CNT does not reach the level of $RuO_2$ catalysts, it performs notably well among similar types of bifunctional non-precious metal catalysts. One significant reason is the high input amount required for $RuO_2$. When normalized to the amount of metal species input, the OER performance of CoFe/CNT is considerably superior to that of $RuO_2$. Similar to the trends observed during ORR catalysis, FeNC, and CoNC require overpotentials of 394 and 392 mV, respectively, to achieve a current density of 10 mA $cm^{-2}$ in OER catalysis. However, in terms of kinetic rates, as reflected in the OER Tafel slopes, the difference between CoFe/CNT and $RuO_2$, as well as FeNC and CoNC, is evident. CoFe/CNT performs slightly better, yet all surpass NC in terms of performance. After 24 h of stability testing, the overpotential driven by CoFe/CNT at a current density of 10 mA $cm^{-2}$ showed virtually no change. These results demonstrate that CoFe/CNT also possesses catalytic capability for the OER, making it an effective bifunctional catalyst compared with other

 

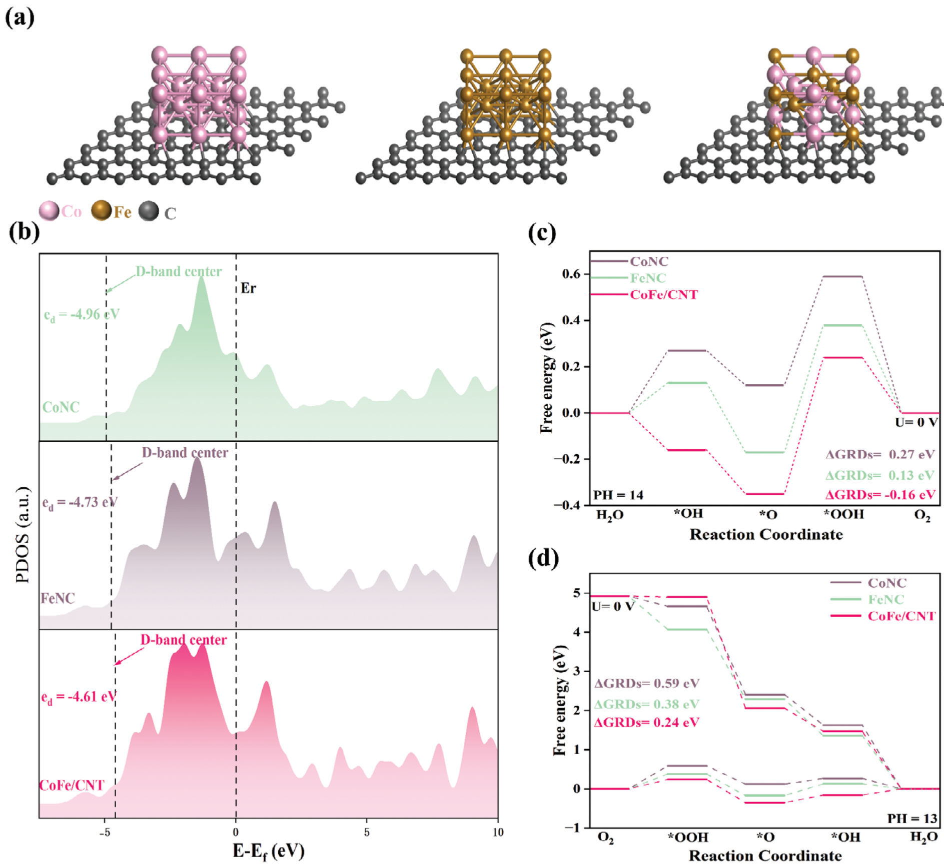


**Figure 5.** CoNC, FeNC, and CoFe/CNT: a) Theoretical models. b) Projected density of states (PDOS). c) Free energy diagram for the OER. d) Free energy diagram for ORR.

bifunctional catalysts, as shown in Figure 4i. All these factors indicate that CoFe/CNT is an advanced catalyst suitable for ORR and OER, exhibiting outstanding performance compared with similar catalysts (refer to Table S2, Supporting Information).

### 2.3. Mechanism Analysis for Enhanced Electrochemical Performance

**Figure 5a** illustrates the theoretical models of various catalysts; specific modeling details are shown in Figure S14 (Supporting Information), and Figure 5b presents the projected density of states (PDOS) analysis. The results show that the d-band center of CoFe/CNT is located at −4.61 eV, closer to the Fermi level, compared with that of CoNC (−4.96 eV) and FeNC (−4.73 eV). This shift is crucial for improving the adsorption of intermediates such as $*O_2$ and *OH, which is consistent with the Ultraviolet Photoelectron Spectroscopy (UPS) results. The PDOS analysis also revealed a substantial overlap between the Fe and Co d-orbitals in CoFe/CNT, optimizing the electron density at the catalyst surface. This structural modification not only enhances the adsorption of oxygen molecules and intermediates but also lowers the activation energy of the reactants, thus reducing the energy barriers for both the OER and ORR. Specifically, CoFe/CNT demonstrates stronger oxygen adsorption, particularly during the $*O_2$ generation and conversion steps, which is essential for improving battery efficiency.

To further investigate the reaction mechanisms and electrocatalytic performance of CoFe/CNT in the OER and ORR, Gibbs free energy calculations were performed, as shown in Figure 5c,d. In the OER, the rate-determining step typically involves the conversion of water into *OH intermediates. CoFe/CNT exhibited the lowest reaction barrier (−0.16 eV), which was significantly lower than those of the other catalysts, indicating a reduced OER overpotential. This lowered reaction barrier enabled CoFe/CNT to facilitate the oxidation of water more effectively, minimizing energy consumption in the OER process. The rate-determining step in the ORR is the adsorption and activation of $O_2$ on the catalyst surface. CoFe/CNT shows a reaction barrier of 0.24 eV, much lower than that of FeNC (0.38 eV) and CoNC (0.59 eV), demonstrating superior kinetics in ORR. This lower reaction barrier allows CoFe/CNT to adsorb oxygen more efficiently and promotes its conversion into reaction intermediates, which is critical for accelerating the ORR rate. In summary, the electronic structure optimization of CoFe/CNT, particularly the shift in the d-band center and the synergistic effects between Fe and Co, endows it with exceptional catalytic performance in ZABs.

### 2.4. Performance of ZABs

To further evaluate the practical applications of the electrocatalyst, bifunctional CoFe/CNT was used as the cathode in a

 

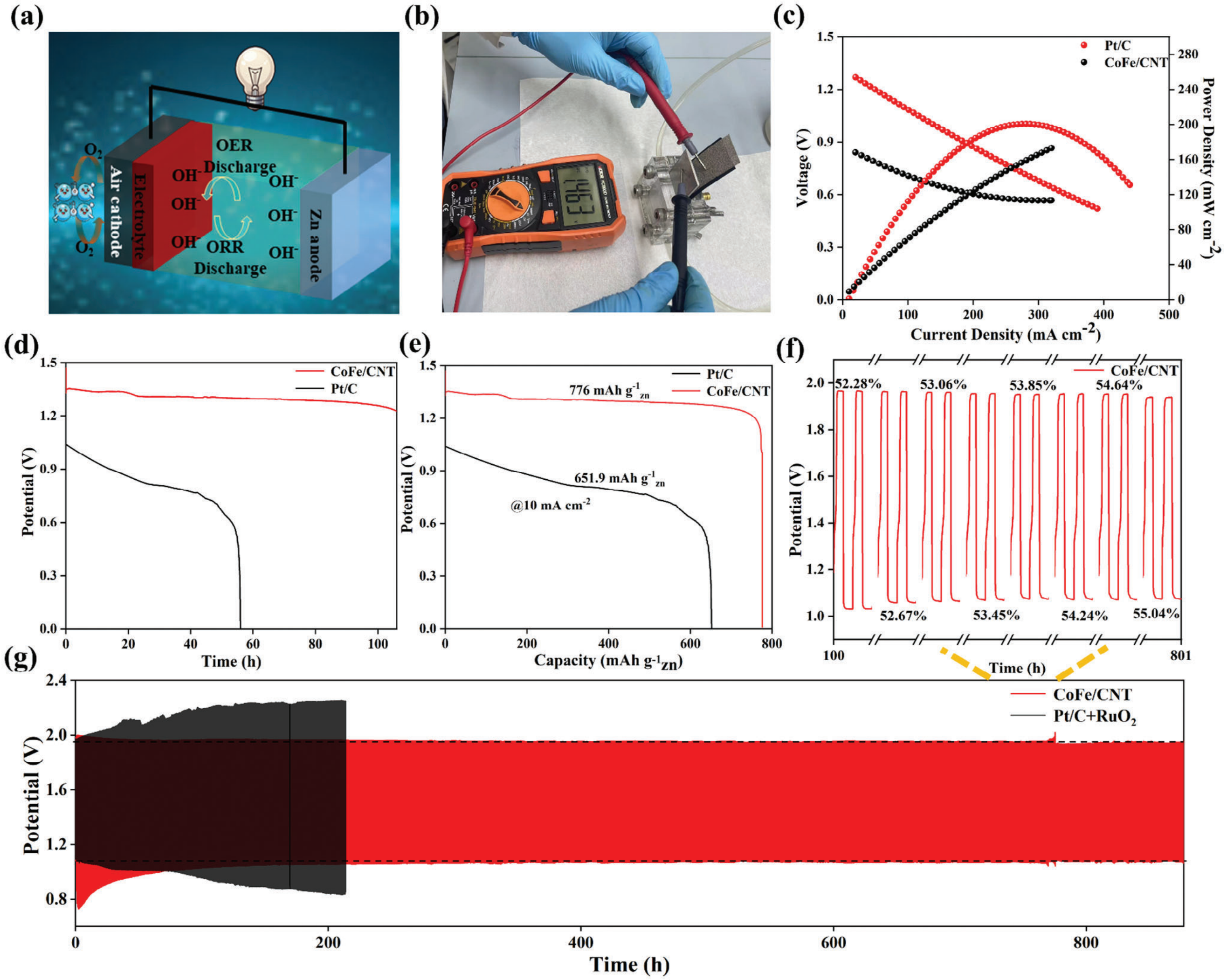


**Figure 6.** Performance and testing of rechargeable ZABs: a) schematic of the working principle of rechargeable ZABs, b) photograph of the battery assembly and voltage measurement, c) current density–voltage curves for CoFe/CNT and Pt/C electrodes, d) discharge voltage versus time curves, e) discharge capacity curves, f) cycling charge–discharge curves for small-area ZABs, and g) charge–discharge cycling test.

domestically produced primary ZAB. Its basic composition is shown in **Figure 6**a. For comparison, a Pt/C-based ZAB was also prepared and tested using the same parameters. Additionally, the battery with the CoFe/CNT air cathode exhibited a high open-circuit voltage of 1.463 V, as shown in Figure 6b. As shown in Figure 6c, the CoFe/CNT-based ZAB demonstrated a peak power density of 220 mW $cm^{-2}$ at high current densities, surpassing the 165.1 mW $cm^{-2}$ of the Pt/C catalyst in the same ZAB setup. Moreover, the discharge curve of the ZAB at a current density of 10 mA $cm^{-2}$ is shown in Figure 6d. The initial voltage of the CoFe/CNT-based battery was high, reaching 1.32 V, which is greater than the 1.04 V of the Pt/C-based battery. Notably, the CoFe/CNT-based ZAB continued to discharge for 106 h while maintaining a voltage of 1.22 V. By contrast, the Pt/C-based battery discharged for only 55 h before experiencing a significant voltage drop. As shown in Figure 6e, the capacity of the CoFe/CNT-based ZAB reached 776 mAh $g^{-1}$ at 10 mA $cm^{-2}$, significantly surpassing the 651.9 mAh $g^{-1}$ of Pt/C and outperforming other reported electrocatalysts.[68–70] To explore the OER performance of CoFe/CNTs, rechargeable ZABs were prepared using the same manufacturing process as that for the primary cells, as shown in Figure 6f,g. The only difference was the addition of 0.2 M zinc acetate to the original 6.0 M potassium hydroxide electrolyte.[71] Figure 6f,g shows the discharge/charge polarization curves of the ZABs with CoFe/CNT and Pt/C/$RuO_2$ cathodes. It is evident that the CoFe/CNT-based battery exhibited much smaller discharge–charge voltage gaps at the same current density compared with those of the commercial Pt/C/$RuO_2$-based battery. The stability of the CoFe/CNT-based ZAB was evaluated over an extended cycling period, as shown in Figure 6f. The voltage profile of the battery exhibited stable charge–discharge cycles between 1.0 V and ≈2.0 V, demonstrating robust electrochemical performance. The efficiency for each cycle was calculated by determining the ratio of the discharge plateau voltage to the charge plateau voltage. Over time, the battery efficiency increased progressively from 52.28% at 100 h to 55.04% at 870 h. This improvement in efficiency highlights the potential of this material for long-term applications. The observed data indicate that CoFe/CNT composites can provide sustained and reliable performance over multiple charge–discharge cycles, making them promising candidates for high-efficiency ZABs. Practical tests using ZABs demonstrated that even after 1750 charge–discharge cycles, the charging and discharging voltages of CoFe/CNT did not significantly decrease from the initial state, as shown in Figure 6g. By contrast, the ZAB with a Pt/C+$RuO_2$ air cathode showed significant voltage decay after ≈420 cycles. Thus, the stability of CoFe/CNT in catalyzing the ORR and OER in the ZAB devices was verified. Practical tests using ZABs demonstrated the application potential of CoFe/CNT as a bifunctional catalyst,[17,72] with a performance superior to those of similar catalysts (Table S3, Supporting Information). This demonstrates that CoFe/CNT is a highly effective catalyst for ZABs, reflecting its robustness and enhanced electrochemical capabilities in energy-conversion applications.

## 3. Conclusion

In this paper, we introduce a novel concept that combines electronic pumping with alloy crystal facet engineering, thereby providing a new perspective for promoting electron enrichment through material design. At the nanoscale, electronic pumping leverages the chemical potential difference to drive electron migration between regions, separating and transferring electron–hole pairs, thereby accelerating reaction kinetics and improving reaction rates. When applied to alloy materials, particularly those with varied exposed crystal facets, engineering these facets can optimize the electron transport pathways and enhance the catalytic performance. Systematic experimental measurements, in situ and ex situ studies, and DFT calculations confirmed that the electron transport behavior differs significantly across crystal facets. Some facets are more conducive to rapid electron conduction, whereas others may trap or scatter electrons, hindering their migration speed and distance. Shorter diffusion paths result in lower energy losses and enable more electrons to reach the active sites for the reaction. In practical applications, ZABs using this electrocatalyst exhibit outstanding performance, with a peak power density of 220 mW $cm^{-2}$, a discharge capacity of 776 mA h $g^{-1}$, and excellent cycling stability. Control experiments confirmed that CoFe alloy particles were the primary active sites for the ORR, enhanced by nitrogen-doped carbon, and affirmed their dual functionality for the OER. This study underscores the potential of this cost-effective electrocatalyst for advancing ZAB technology.

## 4. Experimental Section

*Chemicals and Materials*: Cobalt(II) acetate tetrahydrate ($(CH_3COO)_2Co{\cdot}4H_2O$, 99.99%, Aladdin), iron(III) nitrate nonahydrate ($Fe(NO_3)_3{\cdot}9H_2O$, 98%, Aladdin), zinc nitrate hexahydrate ($Zn(NO_3)_2{\cdot}6H_2O$, 99.9%, Aladdin), 2-methylimidazole ($C_4H_6N_2$, 99%, Aladdin), and potassium hydroxide (KOH, 99.99%, Aladdin) were used for the synthesis of catalysts and electrocatalytic testing. High-purity graphite rods, glassy carbon electrodes, and Hg/HgO electrodes filled with a 1 M KOH solution were also employed. Nafion (5 wt.%) solution was purchased from Macklin. A commercial 20% Pt/C catalyst was obtained from Johnson Matthey and platinum disk electrodes were sourced from Tianjin Aidahengsheng Technology Co., Ltd. High-purity Zn sheets, waterproof breathable membranes, and ZABs components were obtained from Changsha Puslin (China). Additionally, ultrapure water with a resistivity of 18.25 MΩ·cm was used. All chemicals and reagents were obtained from commercial sources and used without further purification.

*Synthesis of ZIF-8-Co/Fe*: 3.24 g 2-methylimidazole was added to methanol (80 mL). After stirring for 5 min, 2.94 g of $Zn(NO_3)_2{\cdot}6H_2O$ was added and the mixture was stirred for 30 min. Subsequently, 0.12 g of $(CH_3COO)_2Co{\cdot}4H_2O$ and 0.12 g of $Fe(NO_3)3{\cdot}9H_2O$ were added to the solution and stirred overnight. The precipitate was collected by centrifugation and washed five times with deionized water and anhydrous ethanol, followed by drying under vacuum at 60 °C for 24 h. Finally, the obtained product was mechanically milled for 30 min. The synthesis methods for the other samples, such as FeNC, CoNC, and NC, were nearly identical, except that no corresponding metal sources were added, and certain steps were omitted.

*Synthesis of CoFe/CNT*: 0.1 g of ZIF-8-Co/Fe nanododecahedra was placed in a quartz boat and then positioned at the center of a tube furnace. The system was heated to 600 °C and held for 1 h under a nitrogen atmosphere, and then heated to 920 °C and maintained for 2 h at a heating rate of 5 °C $min^{-1}$. After heat treatment, the sample was cooled to room temperature under a nitrogen gas flow. The same procedure was followed for the preparation of the other samples.

## Supporting Information

Supporting Information is available from the Wiley Online Library or from the author.

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## Conflict of Interest

The authors declare no conflict of interest.

## Data Availability Statement

The data that support the findings of this study are available from the corresponding author upon reasonable request.

## Keywords

bifunctional electrocatalyst, CoFe nanoparticles, multiwalled carbon nanotubes, ZIF-8, zinc–air batteries

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